

# CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

Tuong-Hung Cao<sup>[0009-0003-5886-2666]</sup>, Ngoc-Anh Le<sup>[0009-0006-7453-1009]</sup>, Nhu-Quynh Nguyen<sup>[0009-0008-7396-697X]</sup>, Minh-Chanh Trang<sup>[0009-0004-7446-522X]</sup>, and Lan Phuong Phan

College of Information and Communication Technology,  
Can Tho University, Can Tho, Vietnam  
{hungb2303873, anhb2303861, quynhb2303777,  
chanhb2303864}@student.ctu.edu.vn  
pplan@ctu.edu.vn

**Abstract.** University counseling spans relational curricula, cohort-specific fees, exemption policies, and narrative regulations, each requiring distinct evidence representations. Whether decomposing this heterogeneous task into specialized agents improves reliability over an equivalent single agent remains underexplored. We present CTU-Chat, a supervisor-routed multi-agent system assigning queries to four bounded specialists with scoped tools and representation-specific evidence paths: Neo4j graph traversal for curricula, structured lookup with deterministic calculation for financial data, and hybrid retrieval with neural reranking for regulations. We evaluate CTU-Chat across three scenarios on a 100-query held-out benchmark: (1) architecture-level comparison against a matched single-agent baseline and a routed generic variant under identical capabilities; (2) architectural ablations removing specialist policies, tool isolation, heterogeneous evidence, deterministic calculation, and route repair; and (3) a semantic-neighbor tool-ambiguity stress test across 11, 31, and 51 tools. In architecture comparison, CTU-Chat achieves a 10.0 pp higher Automated Fact Pass Rate ( $\text{Pass}_{\geq 50\%}$ ) point estimate (76.0% vs. 66.0%; paired 95% CI [+2.0, +19.0] pp; exact McNemar  $p=0.0414$ ) with 41.4% fewer recorded specialist input tokens, alongside an unlogged upstream routing overhead of  $\sim 650$  tokens and sequential dispatch latency (10.65 s pipeline vs. 5.16 s post-retrieval baseline). Under registry expansion to 51 tools, the scoped specialist sustains 88.0% tool E2E Success (vs. 78.7% for Top-10 selection). These findings indicate that domain-scoped specialization and heterogeneous evidence allocation support institutional counseling reliability, with measurable trade-offs in coordination latency and token footprint.

**Keywords:** Multi-Agent Systems · Supervisor Routing · Architectural Ablation · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

## 1 Introduction

Can Tho University (CTU) serves nearly 49,000 undergraduates in 2025–2026 [1]. Its scale and evolving regulations create advisory needs spanning heterogeneous knowledge

forms: (1) **academic curricula** encoding hierarchies and prerequisites; (2) **tuition schedules** varying by cohort and field; (3) **exemption and scholarship policies** combining eligibility conditions with rates; and (4) **administrative regulations** comprising lengthy narrative documents. Prior systems address individual domains—CAAS for IT admission counseling [10] and REBot for regulation advising [9]—but a single representation or retrieval method may be unsuitable across all domains.

RAG and GraphRAG augment LLMs with documents and knowledge graphs [8,5,3], while agentic architectures support tool use and multi-agent coordination [20,18,16]. These approaches do not prescribe how heterogeneous institutional evidence should be represented, routed, and processed. Known limitations in language-model arithmetic [14] further motivate deterministic calculation for financial queries.

**Central hypothesis.** When a counseling system must handle diverse knowledge forms and semantically adjacent tools, routing queries to bounded specialists with representation-specific evidence paths may improve reliability relative to a single agent with equivalent capabilities, albeit with coordination latency. Whether this trade-off is favorable requires direct comparative evaluation.

These challenges motivate three gaps: (1) limited evidence comparing multi-agent and single-agent architectures under matched capabilities; (2) limited mechanistic understanding of which components drive observed differences; and (3) limited evidence on how bounded specialization maintains reliability as the tool registry grows. We formulate three research questions:

- **RQ1:** How does the proposed multi-agent architecture compare with a functionally matched single-agent in end-to-end reliability, evidence grounding, and cost?
- **RQ2:** What performance changes are observed when specialist policies, tool isolation, representation-specific evidence access, deterministic computation, and route repair are individually removed or replaced?
- **RQ3:** How does bounded specialist routing affect tool-selection reliability as semantically overlapping tools expand the registry from 11 to 51 tools?

This work provides three contributions: **C1**—a supervisor-routed bounded multi-agent architecture with four specialists, deterministic route repair, and partitioned tool privileges; **C2**—an architecture-level comparative evaluation against a functionally matched single-agent baseline and a routed generic-worker variant; and **C3**—architectural ablations of five components and a semantic-neighbor tool-registry stress test across 11–51 tools.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the CTU-Chat architecture. Section 4 describes the experimental setup. Section 5 reports results. Section 6 discusses findings and limitations. Section 7 concludes.

## 2 Related Work

### 2.1 Vietnamese NLP and Retrieval Foundations

Vietnamese is an isolating, tonal language where white spaces separate syllables rather than words [11]; 85% of word types contain at least two syllables. While domain-adapted

bi-encoders capture semantic paraphrasing, Vietnamese higher-education decrees rely on rigid Sino-Vietnamese legal terminology and alphanumeric cohort codes, motivating coupling dense embeddings with exact lexical matching.

BM25 [15] provides reliable exact token matching for rigid identifiers but cannot capture paraphrases. Dense retrieval [7,8] captures paraphrastic intent but may struggle with rare alphanumeric tokens. Reciprocal Rank Fusion (RRF) [2] merges BM25 and dense candidates without score calibration, while neural cross-encoders [12,19] jointly encode query–passage pairs for reranking. Adaptive-RAG [6] demonstrates complexity-based retrieval selection. These approaches motivate dynamic retrieval but do not allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

## 2.2 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples [13]. GraphRAG [3] leverages graph topology to uncover multi-hop associative paths that escape flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction may introduce spurious relationships, motivating selective graph construction restricted to domains with well-defined relational schemas.

## 2.3 Agentic RAG and Multi-Agent Orchestration

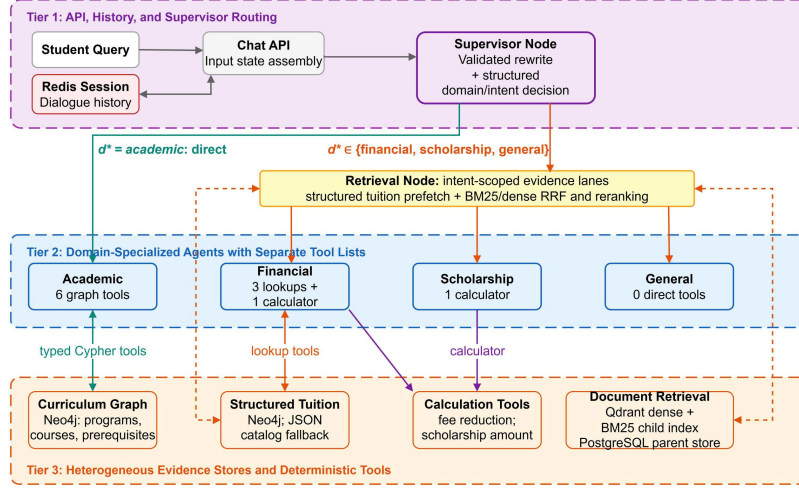
ReAct [20] demonstrates interleaved reasoning and action, AutoGen [18] provides a multi-agent conversation framework, and Toolformer [16] studies tool invocation. For institutional counseling, these capabilities raise design risks without explicit constraints: broad tool exposure increases errors; peer-to-peer exchanges increase coordination cost; a shared index mixes evidence across domains. Controlled comparisons against functionally matched single-agent baselines remain limited, as most evaluations report absolute performance without isolating the contribution of multi-agent decomposition from underlying capabilities.

## 2.4 Educational Counseling Systems

Educational chatbots assist in advising and learning support [17]. CAAS supports admission counseling for IT programs [10], while REBot combines retrieval with category-guided GraphRAG for academic regulations [9]. CTU-CHAT departs from prior systems across three aspects: (1) *Task scope*: modeling 113 curricula as relational property graphs with deterministic arithmetic for tuition; (2) *Multi-agent routing*: a LangGraph supervisor routes to four bounded specialists with isolated evidence spaces; and (3) *Architectural evaluation*: direct comparison against functionally matched baselines with component-level ablations.

## 2.5 Research Gaps

Synthesizing the reviewed approaches identifies three gaps: (1) limited architecture-level evidence comparing multi-agent and single-agent systems under matched capabilities;



**Fig. 1.** Implemented CTU-CHAT architecture. Redis history enters the graph state; the supervisor’s structured decision is contract-checked before conditional dispatch. Academic queries enter the graph-enabled agent; other domains pass through intent-scoped retrieval.

(2) limited mechanistic understanding of which components drive observed differences; and (3) limited robustness evidence under tool ambiguity as semantic neighbors accumulate. These gaps motivate the evaluation design in Sections 4–5.

### 3 Proposed Model

This section presents CTU-CHAT’s architecture, algorithms, and evidence-acquisition mechanisms. The design follows three principles: *centralized agent coordination* (a contract-checked supervisor dispatches each query to one domain agent); *heterogeneous knowledge allocation* (each information structure maps to a corresponding evidence representation); and *representation-aware evidence routing* (graph-tool or intent-scoped retrieval path according to the assigned domain).

#### 3.1 Overall Architecture

Figure 1 illustrates three coordinated tiers: (1) the Interaction and Centralized Supervisor Tier, (2) the Domain-Specialized Multi-Agent Core, and (3) the Heterogeneous Knowledge Storage and Tool Tier.

CTU-CHAT specializes LangGraph through typed state (`AgentState` with validated rewrite, routes, evidence, and response), an intent-owner contract that checks structured supervisor output before dispatch, conditional routing (academic queries enter the graph agent directly; others pass through retrieval), acyclic orchestration (specialists return to END without peer chatter), and asymmetric tool registries (scoping tools per specialist). In live deployment, CTU-CHAT maintains multi-turn session state via Redis and applies

validated query rewriting. For the offline experimental evaluation in Section 4, the system is evaluated in a controlled single-turn benchmark harness with fixed queries, isolating the routing, retrieval, and tool-scoping mechanisms without query-rewriting confounds.

### 3.2 Neo4j Knowledge Graph and Cypher Tools

The Neo4j graph is built from 114 Markdown source files covering 111 undergraduate curricula and fee schedules. The resulting graph stores approximately 7,000 nodes across 10 labels: curriculum entities (e.g., Program, Course, Block, PLO, PEO, Track, Job) and tuition entities (TuitionFee, TuitionPolicy, ExemptionRate) connected by over 16,000 edges across 13 typed relationships encoding containment, prerequisites, program outcomes, tracks, occupations, tuition, and exemption bases.

The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies arguments. Each tool normalizes identifiers and calls a fixed, parameterized Cypher query implementing three query classes: *attribute lookup/search* (program codes, names, comparisons), *structural traversal* (prerequisite chains up to ten hops, course intersection, reverse path search), and *structured financial lookup* (tuition, policies, exemption rates). This design binds agent actions to deterministic graph operations, reducing exposure to errors from unconstrained Cypher generation.

### 3.3 Routing and Deterministic Arithmetic

For a query  $q$  and its dialogue predecessor, the supervisor first attempts a validated standalone rewrite, then predicts domain  $d_{\text{raw}}$  and intent  $\iota_{\text{raw}}$  under a structured schema, falling back to *general/other* on parsing failure. The ownership contract repairs rule conflicts or incompatible domain–intent pairs. Academic queries proceed directly to the graph agent; other domains build intent-scoped retrieval lanes, with financial queries first attempting Neo4j tuition lookup and then JSON fallback.

**Deterministic Tuition Arithmetic.** For fee-reduction requests, the tool `tinh_toan_hoc_phi` receives three parameters—per-credit tuition  $T$ , per-credit exemption basis  $B$ , and percentage  $p$ —and applies:

$$D = B \frac{p}{100}, \quad P = \max(0, T - D). \quad (1)$$

The tool enforces  $0 \leq p \leq 100$  and clamps  $P$  at zero. The contribution lies in composing validated rewriting, constrained routing, and deterministic arithmetic within the multi-agent workflow.

### 3.4 Heterogeneous Knowledge Allocation and Hybrid Retrieval

Rather than a uniform document index, CTU-CHAT preserves relational curricula in Neo4j, accesses cohort-dependent tuition through structured lookup, and retains narrative regulations in document retrieval. Academic queries enter the graph agent directly; others enter retrieval. Financial retrieval first attempts Neo4j tuition lookup with JSON fallback; the repaired intent then selects metadata-filtered document lanes.

For narrative regulations and scholarship criteria, BM25 and Qdrant search 400-character child chunks; parent sections (2,800-char target, 100-char overlap) are loaded from PostgreSQL and fused via RRF:  $s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25}, \text{dense}\}} (60 + r_s(d))^{-1}$ . The full stack pools intent-governed and unfiltered candidates, deduplicates, and reranks with `bge-reranker-v2-m3`. Scores below 0.20 are removed unless fewer than two candidates remain; graph evidence is added under fixed quotas and up to seven passages are packed for generation.

### 3.5 Cross-Domain Query Handling

A bounded specialist architecture raises the question of how queries spanning multiple domains are handled. CTU-CHAT adopts a *single-owner with signal-based evidence prefetch* policy: the supervisor selects one primary owner based on dominant intent, and the retrieval layer expands evidence lanes when keyword signals indicate related domains—for example, adding academic-program or academic-regulation lanes when curriculum terms are detected, a tuition lane when fee-related terms appear, or a scholarship lane when scholarship keywords are present. Specialists do not invoke other specialists; if the query requires evidence outside the owner’s scope, the signal-detected lanes prefetch it through retrieval rather than cross-specialist delegation. This design trades full cross-domain tool composability for bounded execution with targeted evidence expansion. The 20 cross-domain held-out queries directly test this mechanism.

## 4 Experimental Setup

### 4.1 Dataset and Annotation

We evaluate CTU-CHAT on 200 administrative queries constructed through a document-grounded protocol over official CTU statutes. The 100-query development split supports index calibration and prompt engineering. The 100-query held-out set follows a predefined distribution: 40 direct, 20 within-domain multi-hop, 20 cross-domain, 10 entity-comparison, 5 temporal, and 5 adversarial queries (51 formal, 49 colloquial; 38 single-source, 62 multi-source). It contains 19 Academic, 28 Financial, 17 Scholarship, and 16 General single-domain queries; the remaining 20 combine evidence across domains. Candidate queries and ground-truth evidence were extracted from official statutes: one annotator established reference answers, required facts, and gold sources, while a second annotator independently verified all annotations; disagreements were adjudicated against official regulatory text. An audit found no cross-split duplicates (max 5-gram Jaccard = 0.1875). Each query is annotated along nine dimensions, including domain, evidence type, and tool requirements, to support fine-grained failure analysis.

### 4.2 Baselines and Configurations

**Fairness principle and execution contract.** For Scenario 1, all architecture configurations share the same LLM (Gemini 2.5 Flash Lite,  $T=0.0$ ), knowledge representations, retrieval stack (BM25 + dense + RRF + `bge-reranker-v2-m3`,  $k=7$ ), and deterministic calculator

access. The intended contrast is the *orchestration architecture*. Scenario 2 intentionally changes one component at a time. To reduce retrieval variance in Scenario 1, all three architecture variants evaluate over the identical shared full-stack evidence retrieval output, yielding matched pre-generation Source Recall (78.6%) and Source AP (70.8%). The shared evidence context excludes outputs of tools whose invocation is evaluated; tool-derived results are available only after successful tool execution. Differences in final answers may be associated with the combined changes in tool privilege scoping, prompt specialization, and agent routing; Scenario 1 does not isolate their individual effects. Multi-agent variants use intent-routed scoped registries with identical execution bounds: maximum 3 tool calls, matching context limits, 2,048 output-token budget, and 30-second timeouts. Token accounting in the offline harness records specialist agent execution; supervisor routing introduces an estimated upstream overhead of  $\sim 650$  input tokens per query (derived from the supervisor prompt template, structured routing schema, and average query length) in multi-agent variants. Reported input and output token comparisons between S1-C and the single-agent baseline reflect specialist agent context scoping, and are explicitly qualified alongside total pipeline costs.

*Scenario 1 — Architecture Comparison (RQ1):* Three configurations are evaluated on the held-out set ( $N=100$  queries, 300 runs under greedy decoding  $T=0.0$ ): **S1-A:** Unified Single Agent with all 11 tools, no supervisor or routing; **S1-B:** Routed Generic Worker—supervisor routes to domain workers with scoped tools but a single generic prompt; **S1-C:** Full CTU-Chat with supervisor, contract repair, specialist prompts, scoped registries, and deterministic computation.

*Scenario 2 — Ablation (RQ2):* Five ablations each remove one component from Full CTU-Chat ( $N=100$  queries per ablation,  $T=0.0$ ): **S2-A1:** No Specialist Policy (generic worker prompt); **S2-A2:** No Tool Isolation (all 11 tools visible); **S2-A3:** Uniform Text-RAG Baseline (graph data serialized to text, traversal disabled); **S2-A4:** No Deterministic Calculation (LLM generates arithmetic directly); **S2-A5:** No Route Repair (raw supervisor output used directly).

*Scenario 3 — Tool Ambiguity (RQ3):* Fifty fixed cases are evaluated while 40 semantic-neighbor tools expand registries to 11, 31, and 51 tools. Four configurations are compared: Full-Registry Single Agent, Global Top-10 Single Agent, Routed Generic, and Routed Specialist. Each case uses 3 counterbalanced tool-order blocks, yielding  $50 \times 4 \times 3 \times 3 = 1,800$  decisions across counterbalanced blocks.

### 4.3 Evaluation Metrics

Metrics span four groups. *Final-answer quality:* Required-Fact Coverage (continuous proportion of reference facts matched via substring containment in generated response), Answer Correctness, Answer Relevancy, and Faithfulness (Ragas [4]). *Evidence quality:* Source Recall, Source AP, Context Recall, Context Precision. *Execution reliability:* Tool Invocations per Query, Tool Selection Accuracy, Argument EM/Contract Match, Result Accuracy, and route repair rate. *Operational cost:* input/output tokens, LLM call

counts, and wall-clock latency. Source Recall and Source AP serve as controlled retrieval diagnostics, identical by design across Scenario 1 configurations.

**Primary Endpoint Definitions.** In Scenarios 1 and 2, the primary decision endpoint is the *Automated Fact Pass Rate* ( $\text{Pass}_{\geq 50\%}$ ), defined programmatically as the proportion of responses where at least 50% of predefined required fact phrases are matched via automated substring containment ( $\text{FactCov} \geq 0.50$ ), complemented by continuous Fact Coverage. Because this lexical metric evaluates surface phrase occurrence rather than semantic validity or arithmetic computation, it serves as a surface-grounding proxy and is complemented by independent deterministic evaluations: in Scenario 3 by *Tool E2E Success* (correct tool selection, valid parameters, and error-free execution), and on the arithmetic subset ( $N=9$ ) by numeric exact-match auditing against official bursar schedules.

#### 4.4 Statistical Protocol and Measurement Scope

Evaluations use greedy decoding ( $T=0.0$ ) to minimize sampling variance. Paired differences across configurations are computed at the query level ( $N=100$ ); 95% CIs are estimated using 10,000 bootstrap resamples. The primary confirmatory hypothesis compares CTU-Chat versus Unified Single Agent on  $\text{Pass}_{\geq 50\%}$ . Timing follows configuration boundaries: baseline S1-A latency measures agent invocation post-retrieval, while multi-agent variants measure pipeline latency inclusive of supervisor routing, route repair, and specialist execution. Reported tokens record specialist agent execution; upstream supervisor routing adds an estimated overhead of  $\sim 650$  input tokens per query (derived from prompt and schema length).

#### 4.5 Infrastructure

Gemini 2.5 Flash Lite via Vertex AI ( $T=0.0$ ). Neural reranking on a GTX 1650 (evaluation hardware). Neo4j 5.20 stores relational curricula; PostgreSQL stores parent chunks (2,800-char, 100 overlap); Qdrant indexes child chunks (400-char, 50 overlap) with a 768-dimensional vietnamese-bi-encoder. System uses `bge-reranker-v2-m3`, LangGraph StateGraph, and Redis 7. Latency excludes the first warm-up run. Prompt templates, tool contracts, and graph schema are archived.

## 5 Results

### 5.1 Scenario 1 — Architecture-Level Comparison

Table 1 compares the three configurations on the 100-query held-out benchmark. Because pre-generation evidence retrieval is held fixed across configurations, Source Recall (78.6%) and Source AP (70.8%) are identical across all three architectures. The observed differences in final answer quality and operational cost may be associated with the combined orchestration changes described in Section 4.

**Pairwise bootstrap analysis** (10,000 resamples): (1) CTU-Chat vs. Single Agent: CTU-Chat achieves a 10.0pp higher Automated Fact Pass Rate ( $\text{Pass}_{\geq 50\%}$ ) point estimate (76.0% vs. 66.0%, paired 95% bootstrap CI  $[+2.0, +19.0]$  pp; exact McNemar



**Table 1.** Scenario 1: Architecture-level comparison ( $N = 100$ ). Pass<sub>50</sub> = Automated Fact Pass Rate ( $\geq 50\%$  required facts). Tools/Q = mean tool invocations ( $\downarrow$ ). In/Out Tok = specialist agent tokens. Bold = best reliability / lowest cost.

| Arch.           | Pass <sub>50</sub> | Fact        | Src.R | Src.AP | Tools/Q     | In Tok      | Out Tok     | Lat. (ms)   |
|-----------------|--------------------|-------------|-------|--------|-------------|-------------|-------------|-------------|
| Single Agent    | 66.0               | 57.5        | 78.6  | 70.8   | 0.41        | 6296        | 1530        | <b>5159</b> |
| Routed Gen.     | 66.0               | 54.2        | 78.6  | 70.8   | 0.31        | <b>3473</b> | 1415        | 8921        |
| <b>CTU-Chat</b> | <b>76.0</b>        | <b>61.0</b> | 78.6  | 70.8   | <b>0.28</b> | 3690        | <b>2092</b> | 10652       |

$p=0.0414$ ), alongside higher Fact Coverage ( $\Delta = +3.58$  pp, 95% CI  $[-1.2, +8.5]$  pp) and 41.4% fewer specialist input tokens (26.1% fewer total specialist tokens:  $\Delta = -2,044$ , CI  $[-2,710, -1,378]$ ; Single: 7,826 vs. CTU: 5,782; supervisor adds  $\sim 650$  input tokens/query), under differing measurement boundaries (10,652 ms end-to-end multi-agent pipeline vs. 5,159 ms post-retrieval single-agent generation). (2) CTU-Chat vs. Routed Generic: specialist policies raise Fact Coverage by +6.85 pp (CI  $[+0.7, +13.0]$  pp,  $p < 0.05$ ). (3) Routed Generic vs. Single Agent: comparable token savings ( $\Delta = -2,824$  input tokens) but a  $-3.27$  pp Fact Coverage point estimate (CI  $[-9.0, +2.4]$  pp); the interval includes zero.

On the 20 cross-domain queries, CTU-CHAT achieves an 80.0% pass rate (vs. 70.0% Single, 65.0% Generic). Threshold sensitivity confirms stability: CTU-CHAT leads significantly at  $\theta \in \{0.40, 0.50\}$  (+10.0 pp, exact McNemar  $p=0.0414$ , 95% CI  $[+1.0, +19.0]$  pp) and leads at  $\theta=0.60$  (54.0% vs. 50.0%). At  $\theta \geq 0.70$ , pass rates compress across architectures; Single Agent’s nominal 2–3 query advantage (e.g., 30.0% vs. 27.0% at  $\theta=0.75$ ) is statistically indistinguishable from noise (McNemar  $p=0.5488$ , 95% CI  $[-10.0, +3.0]$  pp).

**Latency and Token Considerations:** Across configurations, latency reflects distinct operational boundaries: Single Agent (5,159 ms) measures inference post-retrieval, whereas multi-agent variants (8,921 ms for Routed Generic, and 10,652 ms for CTU-CHAT) measure pipeline execution inclusive of routing and contract dispatch. Within multi-agent execution, the +1,731 ms delta between Routed Generic and CTU-CHAT corresponds to specialist prompt instructions producing +677 additional output tokens (2,092 vs. 1,415) for domain-tailored reasoning and evidence formatting. In ablation S2-A4, the system showed 4,540 ms lower latency when deterministic calculators were disabled. Regarding token usage, specialist prompt scoping reduces recorded agent input tokens (3,690 vs. 6,296 in Table 1), while supervisor routing introduces an estimated upstream overhead of  $\sim 650$  input tokens per query (derived from prompt and schema length).

## 5.2 Scenario 2 — Architectural Ablation

Table 2 removes one mechanism at a time from Full CTU-Chat ( $N=100$ ). Deltas are relative to the full system.

The two largest observed degradations occur in *No Specialist Policy* ( $\Delta$ Pass<sub>50</sub> =  $-10.0$  pp, 95% CI  $[-19.0, -1.0]$  pp;  $\Delta$ Fact =  $-6.85$  pp, CI  $[-13.0, -0.7]$  pp) and *Uniform Text-RAG Baseline* ( $\Delta$ Pass<sub>50</sub> =  $-9.0$  pp, CI  $[-15.0, -4.0]$  pp;  $\Delta$ Src.R =  $-12.3$  pp, CI

**Table 2.** Scenario 2: Architectural ablation results ( $N = 100$ ). Panel A reports overall benchmark differences ( $\Delta$  relative to Full CTU-CHAT). Panel B reports Automated Fact Pass Rate ( $\text{Pass}_{\geq 50\%}$ , %) across query families.

| <b>Panel A: Overall Benchmark Ablations</b> |                           |                     |                      |                    |                    |
|---------------------------------------------|---------------------------|---------------------|----------------------|--------------------|--------------------|
| <b>Ablated Component</b>                    | $\Delta\text{Pass}_{.50}$ | $\Delta\text{Fact}$ | $\Delta\text{Src.R}$ | $\Delta\text{Tok}$ | $\Delta\text{Lat}$ |
| No Spec. Policy                             | -10.0                     | -6.85               | 0.0                  | -217               | -1731              |
| Full Tool Vis.                              | 0.0                       | +0.93               | 0.0                  | +1270              | +296               |
| Uniform Text-RAG                            | -9.0                      | -2.18               | -12.3                | -300               | -1910              |
| No Route Rep.                               | -2.0                      | -1.27               | 0.0                  | -219               | -1610              |
| No Determ. Calc. <sup>†</sup>               | -4.0                      | -0.93               | 0.0                  | -782               | -4540              |

| <b>Panel B: Fact-Grounded Pass Rate (<math>\text{Pass}_{\geq 50\%}</math>) by Query Family</b> |                                  |                                       |                                     |                                  |
|------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------------|-------------------------------------|----------------------------------|
| <b>Configuration</b>                                                                           | <b>Graph (<math>n=19</math>)</b> | <b>Struct-fin (<math>n=28</math>)</b> | <b>Narr-reg (<math>n=33</math>)</b> | <b>Cross (<math>n=20</math>)</b> |
| <b>Full CTU-Chat</b>                                                                           | 63.2                             | <b>75.0</b>                           | 81.8                                | <b>80.0</b>                      |
| No Spec. Policy                                                                                | 52.6                             | 67.9                                  | 72.7                                | 65.0                             |
| Full Tool Vis.                                                                                 | <b>68.4</b>                      | 67.9                                  | <b>84.8</b>                         | <b>80.0</b>                      |
| Uniform Text-RAG                                                                               | 63.2                             | 60.7                                  | 78.8                                | 60.0                             |
| No Route Rep.                                                                                  | <b>68.4</b>                      | <b>78.6</b>                           | 78.8                                | 65.0                             |

<sup>†</sup>Full benchmark with calculators disabled.  $\Delta\text{Tok}/\Delta\text{Lat}$  in tokens/ms per query.

$[-16.5, -8.3]$  pp). Removing tool isolation causes no change in point estimate ( $\Delta\text{Pass}_{.50} = 0.0$  pp, CI  $[-6.0, +6.0]$  pp) while expanding input tokens by +1,270. Disabling route repair yields  $\Delta\text{Pass}_{.50} = -2.0$  pp (CI  $[-7.0, +3.0]$  pp), disproportionately harming cross-domain queries ( $-15.0$  pp, dropping from 80.0% to 65.0%). Disabling deterministic calculation produces  $\Delta\text{Pass}_{.50} = -4.0$  pp (CI  $[-9.0, 0.0]$  pp) across the benchmark; on the arithmetic tuition subset ( $N=9$ ), exact-match answers drop from 8/9 to 3/9 ( $-55.6$  pp exact match pass) due to unconstrained arithmetic hallucination. Collectively, the ablations distinguish two functional roles: specialist policies, heterogeneous evidence, and deterministic calculation drive accuracy at the current 11-tool scale, while tool isolation (S2-A2) functions as an efficiency mechanism (+1,270 input tokens,  $\Delta\text{Pass}_{.50} = 0.0$  pp) whose accuracy contribution is examined under larger registry scales in Scenario 3 (Section 5.3). Query family breakdown in Panel B confirms that heterogeneous evidence and specialist routing particularly bolster structured financial and cross-domain questions.

### 5.3 Scenario 3 — Tool Ambiguity Stress Test

Table 3 reports the semantic-neighbor experiment across 11–51 tools ( $N=50$ , 3 counter-balanced tool-order blocks, 1,800 primary decisions).

As the registry scales from 11 to 51 tools, the Full-Registry Single Agent changes from 86.0% to 83.3% E2E while input tokens increase by 124%. The Global Top-10 changes from 83.3% to 78.7% E2E, alongside a Gold Recall change from 100.0% to 96.0%. The Routed Specialist records 88.0% E2E at 51 tools with 9.8 visible tools. At the 51-tool endpoint, Routed Specialist has a +9.3 pp E2E difference over Top-10 Single Agent (95% CI  $[+2.0, +17.3]$  pp) and a +4.7 pp point-estimate difference over Full-Registry Single Agent (95% CI  $[-1.3, +11.3]$  pp) with 52.4% fewer input tokens

**Table 3.** Scenario 3: Semantic-neighbor tool ambiguity ( $N = 50$ , 3 counterbalanced tool-order blocks).

| Reg. | Config.             | Vis. | Gold-R       | Sel.        | E2E         | Tok         | Lat         |
|------|---------------------|------|--------------|-------------|-------------|-------------|-------------|
| 11   | Full-reg single     | 11   | –            | <b>94.0</b> | <b>86.0</b> | 1663        | <b>1098</b> |
|      | Top-10 single       | 10   | <b>100.0</b> | 93.3        | 83.3        | 1541        | 1148        |
|      | Routed gen.         | 5.1  | 98.0         | 90.7        | 80.0        | <b>1204</b> | 1889        |
|      | <b>Routed spec.</b> | 5.1  | 98.0         | 93.3        | 85.3        | 1441        | 1983        |
| 31   | Full-reg single     | 31   | –            | 91.3        | 84.7        | 2706        | 1157        |
|      | Top-10 single       | 10   | <b>98.0</b>  | 90.0        | 81.3        | <b>1149</b> | <b>1033</b> |
|      | Routed gen.         | 9.8  | <b>98.0</b>  | 88.0        | 79.3        | 1396        | 1958        |
|      | <b>Routed spec.</b> | 9.8  | <b>98.0</b>  | <b>93.3</b> | <b>88.7</b> | 1830        | 2015        |
| 51   | Full-reg single     | 51   | –            | 92.7        | 83.3        | 3722        | 1289        |
|      | Top-10 single       | 10   | 96.0         | 86.7        | 78.7        | <b>1036</b> | <b>997</b>  |
|      | Routed gen.         | 9.8  | <b>98.0</b>  | 91.3        | 82.7        | 1340        | 1907        |
|      | <b>Routed spec.</b> | 9.8  | <b>98.0</b>  | <b>92.7</b> | <b>88.0</b> | 1770        | 2021        |

(1,770 vs. 3,722). The degradation interaction (Specialist vs. Top-10 high-minus-low difference) is +7.3 pp (95% CI [0.0, +15.3] pp).

Analysis of stage-level timing across the 1,960 decisions reveals that Routed Specialist maintains a consistent latency profile across registry expansions (2,091 ms at 11 tools vs. 2,116 ms at 51 tools). Here, supervisor routing introduces a fixed overhead of 946.5–961.8 ms (~45% of total decision time), tool gating requires 1,051.6–1,079.3 ms, and tool execution takes 71.0–92.8 ms. In contrast, while the Full-Registry Single Agent avoids supervisor dispatch (0 ms routing overhead), its gating latency increases from 1,098.1 ms to 1,272.2 ms as prompt schema tokens scale from 1,663 (11 tools) to 3,722 (51 tools), whereas the Routed Specialist bounds prompt schema tokens between 1,441 and 1,770 across the same expansion.

#### 5.4 Supporting Retrieval Validation

Table 4 validates the retrieval subsystem independently.

**Table 4.** Supporting evaluation: Progressive retrieval component stacking ( $N = 100$ ).

| Config.               | Hit@1        | Hit@3        | P@5          | R@5          | MRR@10       | Lat. (ms) |
|-----------------------|--------------|--------------|--------------|--------------|--------------|-----------|
| BM25 Lexical          | 0.530        | 0.770        | 0.206        | 0.617        | 0.669        | <b>7</b>  |
| Dense Semantic        | 0.460        | 0.650        | 0.176        | 0.524        | 0.560        | 34        |
| Hybrid RRF            | 0.520        | 0.790        | 0.224        | 0.662        | 0.673        | 38        |
| Hybrid + Reranker     | 0.700        | 0.910        | 0.254        | 0.751        | 0.805        | 15,366    |
| <b>Full Retrieval</b> | <b>0.740</b> | <b>0.920</b> | <b>0.302</b> | <b>0.861</b> | <b>0.833</b> | 21,858    |

BM25 records 0.530 Hit@1 and 0.669 MRR@10, compared with 0.460 Hit@1 and 0.560 MRR@10 for dense semantic search. Hybrid RRF achieves 0.790 Hit@3 and 0.520 Hit@1, balancing lexical precision and semantic recall. Adding the cross-encoder reranker (`bge-reranker-v2-m3`) yields the most pronounced individual retrieval gain, raising Hit@1 from 0.520 to 0.700 (+18.0 pp) and MRR@10 to 0.805 at the expense of reranker inference latency. The full retrieval configuration reaches 0.740 Hit@1, 0.861 Recall@5, and 0.833 MRR@10. Because this final configuration incorporates domain-specific evidence components, the difference cannot be attributed to one component alone.

## 5.5 Failure Analysis

Table 5 categorizes error modes across the 58 failure instances (< 50% Fact Coverage) in Scenario 1 (34 for Single Agent, 24 for CTU-Chat), established through manual qualitative review of individual execution traces and intermediate outputs.

**Table 5.** Failure taxonomy across architecture configurations ( $N = 100$ ). Categories are non-mutually exclusive; a single query may exhibit multiple error modes.

| Category                                     | Failure Mode           | S1-A | S1-C | Category                 | Failure Mode          | S1-A                  | S1-C |
|----------------------------------------------|------------------------|------|------|--------------------------|-----------------------|-----------------------|------|
| Routing                                      | Corrected misroute     | N/A  | 1    | Tool                     | Invalid arguments     | 4                     | 1    |
|                                              | Unrecoverable misroute | N/A  | 3    |                          | Wrong-family call     | 3                     | 1    |
| Retrieval                                    | Gold source absent     | 3    | 3    |                          | Unnecessary call      | 5                     | 1    |
|                                              | Cross-domain omission  | 3    | 2    | Generation               | Result ignored        | 2                     | 1    |
|                                              | Reranker demoted       | 7    | 6    |                          | Partial fact omission | 22                    | 14   |
| <b>Total Failed Queries (&lt; 50% Fact):</b> |                        |      |      | <b>S1-A (Single): 34</b> |                       | <b>S1-C (CTU): 24</b> |      |

CTU-Chat reduces overall failure instances from 34 to 24. Tool errors drop from 12 in Single Agent (4 invalid arguments, 3 wrong-family calls, 5 unnecessary invocations) to 3 in CTU-Chat due to scoped registries. Retrieval omissions (3 gold absent, 7 vs. 6 partial demotions) affect both configurations similarly because pre-generation retrieval was matched. In generation, partial fact omissions under complete retrieval ( $sr = 1.0$ ) decrease from 22 in Single Agent to 14 in CTU-Chat, reflecting more structured synthesis in specialist prompt contracts.

## 6 Discussion

### 6.1 Architectural Findings and Operational Trade-offs

**RQ1 (Architecture-Level Effectiveness):** The architecture-level comparison indicates that supervisor-routed multi-agent decomposition achieves a 10.0 pp higher Automated Fact Pass Rate ( $\text{Pass}_{\geq 50\%}$ ) point estimate than the Unified Single Agent (76.0% vs. 66.0%; paired bootstrap CI [+2.0, +19.0] pp; exact McNemar  $p=0.0414$ ), alongside higher Fact

Coverage (+3.58 pp, 95% CI  $[-1.2, +8.5]$  pp) and 41.4% fewer recorded specialist input tokens (3,690 vs. 6,296; supervisor adds  $\sim 650$  routing tokens/query), while end-to-end dispatch latency exceeds post-retrieval generation (10,652 vs. 5,159 ms). Routed Generic records lower Fact Coverage (54.2% vs. 61.0%;  $\Delta = -6.85$  pp,  $p < 0.05$ ), showing that specialist policies contribute substantially to answer fidelity under shared routing.

**RQ2 & RQ3 (Mechanisms, Tool Isolation, and Registry Scaling):** Component ablations reveal two distinct roles. *Accuracy drivers:* Heterogeneous evidence access (S2-A3) improves Source Recall by +12.3 pp overall (+17.8 pp on finance); specialist policies (S2-A1) govern coverage ( $-6.85$  pp Fact Coverage); route repair (S2-A5) provides cross-domain robustness ( $-15.0$  pp); and calculators (S2-A4) achieve 8/9 numeric exact matches on tuition queries, versus 3/9 without calculators. *Scalability and efficiency:* At 11 tools, removing tool isolation (S2-A2) does not alter pass rate ( $\Delta \text{Pass}_{.50} = 0.0$  pp, CI  $[-6.0, +6.0]$  pp) but increases input tokens by +1,270 (+34.4%), showing that tool isolation acts as a cost-containment safeguard rather than an accuracy driver at small scales. Under 51-tool registry expansion (Scenario 3), the Routed Specialist sustains 88.0% tool E2E Success with 9.8 visible tools, compared with 78.7% for Global Top-10 (+9.3 pp, CI  $[+2.0, +17.3]$  pp) and 83.3% for Full-Registry Single Agent (+4.7 pp, CI  $[-1.3, +11.3]$  pp) while using 52.4% fewer input tokens (1,770 vs. 3,722). In tested registries, specialization kept per-agent tool exposure bounded.

**Latency Decomposition and Operational Guidance:** Operational trade-offs reflect distinct boundaries: (1) *Measurement boundaries:* Single-agent timing measures inference post-retrieval (5,159 ms), whereas multi-agent variants measure end-to-end dispatch inclusive of routing (8,921–10,652 ms). (2) *Specialist overhead:* The observed +1,731 ms delta between generic and specialist multi-agent execution coincides with specialist prompt instructions producing +48% additional output tokens (2,092 vs. 1,415) for domain-tailored reasoning, alongside an estimated upstream supervisor routing overhead of  $\sim 650$  input tokens. (3) *Deployment guidance:* Monolithic single-agent designs suit low-latency, single-domain queries with small tool pools ( $\leq 10$ ); supervisor-routed multi-agent architectures are justified when counseling demands heterogeneous evidence representations, deterministic arithmetic fidelity, and tool isolation under expanding registries.

## 6.2 Relation to Prior Work

While ReAct [20] and Toolformer [16] integrate tools within single agents, full-registry single agents suffer prompt expansion as tools scale. Unlike collaborative frameworks (e.g., AutoGen [18]) where conversational chatter compounds latency, CTU-CHAT enforces bounded single-owner dispatch with route repair. Ablation S2-A3 highlights text retrieval limits on relational curricula, corroborating GraphRAG’s rationale for heterogeneous evidence [3]. Unlike Adaptive-RAG’s linguistic complexity branching [6], CTU-CHAT routes queries by institutional data modality.

## 6.3 Threats to Validity

**Internal and Operational:** Evaluations use Gemini 2.5 Flash Lite with greedy decoding ( $\tau=0$ ); serving variations and network transport remain operational confounds. **External:**

Benchmark queries reflect Can Tho University statutes; multi-campus transfer across bilingual or trimester curricula remains future work. Semantic neighbors in Scenario 3 are controlled synthetic constructs. **Construct:** Automated substring matching ( $\text{Pass}_{\geq 50\%}$ ) evaluates lexical phrase occurrence rather than computational validity (e.g., in query HOUT-DIR-FIN-03, regulatory keywords matched 50% coverage despite an incorrect tuition calculation of 192.183M vs. 40M VND); this is mitigated via deterministic tool validation (Scenario 3) and bursar exact-match audits ( $N=9$ ). **Statistical:** Paired bootstrap resampling uses 10,000 iterations ( $N=100$ ); arithmetic exact match ( $N=9$ ) has a small sub-cohort, though the 8/9 vs. 3/9 delta is substantial.

## 7 Conclusion and Future Work

This study evaluated CTU-CHAT, a supervisor-routed multi-agent RAG framework for university counseling, across architecture comparison, ablation, and tool ambiguity stress testing. Three principal findings address the research questions: (1) CTU-CHAT achieves a 10.0 pp higher Automated Fact Pass Rate ( $\text{Pass}_{\geq 50\%}$ ) than matched single-agent baseline (76.0% vs. 66.0%; paired CI [+2.0, +19.0] pp; exact McNemar  $p=0.0414$ ) with 41.4% fewer specialist input tokens; (2) removing specialist policies reduces pass rate by −10.0 pp, while replacing heterogeneous evidence reduces Source Recall by −12.3 pp; and (3) the Routed Specialist sustains 88.0% tool E2E Success at 51 tools (vs. 83.3% and 78.7% for baselines). Domain policies, representation-specific evidence paths, and deterministic calculation contribute to counseling reliability, while scoped tool access provides scalability safeguards under registry expansion. Future work includes reducing dispatch latency via speculative routing, evaluating cross-institutional transfer, and conducting live counseling trials.

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